Credit Risk Scoring

Executive review | German Credit benchmark | Nested-OOF and calibration revision | 7 November 2026

1. Executive summary

A tuned Random Forest ranks the supplied Good/Bad credit-risk labels better than the shallow Decision Tree baseline. The operating threshold is now selected from nested training OOF predictions: hyperparameter selection is repeated inside each outer fold, so each OOF record is excluded from the model-selection process that predicts it. Platt calibration is fitted only to those nested OOF scores and is cross-fitted again when choosing its own operating threshold. The held-out test set remains evaluation-only.

0.809	0.690	0.153	0.17
Test ROC-AUC	Test average precision	Calibrated Brier score	Calibrated cutoff

Data profile

Records	Predictors	Good / bad-risk labels	Bad-risk prevalence	Missing / duplicates
1,000	20 (7 numeric, 13 categorical)	700 / 300	30%; 2.33:1	0 / 0

Target caveat. The source defines Good/Bad credit-risk classes (original labels 1 and 2). The assignment treats Bad as a default/bad-payer proxy, but the archive does not provide verified realised-default outcomes. Results therefore predict the supplied bad-risk label, not measured default.

Model comparison on the untouched 200-record test set

Operating point	Cutoff	ROC-AUC	Avg. precision	FN	FP	Cost / record
Decision Tree · train OOF	0.21	0.720	0.495	10	76	0.630
Random Forest · nested OOF	0.22	0.809	0.690	8	77	0.585
Random Forest · default 0.50	0.50	0.809	0.690	42	6	1.080

The benchmark cost weights are 5 for approving a bad-risk-labelled record and 1 for declining a good-risk-labelled record. Cost per record is a relative score, not monetary savings.

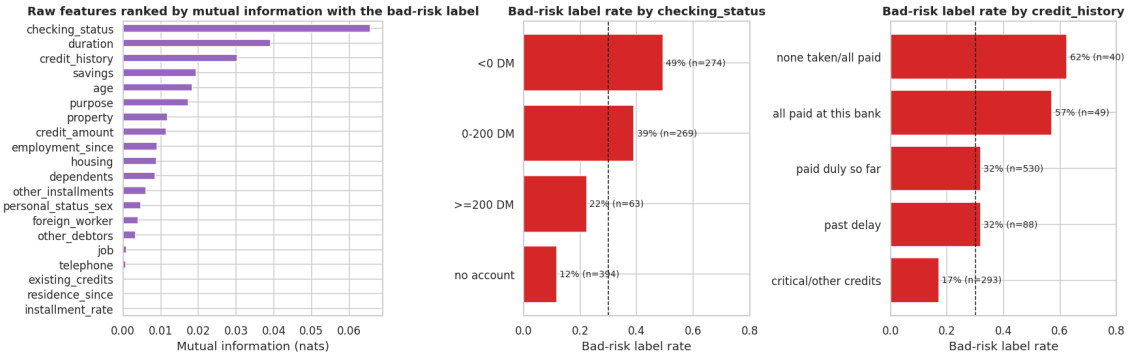


Figure 1. Mutual-information ranking and observed bad-risk-label rates for selected categorical fields. These are descriptive associations, not causal effects.

2. Operating point and probability calibration

Nested OOF threshold selection

The Random Forest uses five outer stratified folds. Within each outer-training partition, a three-fold inner search selects the hyperparameters; the held-out outer fold is then predicted once. The five fold searches selected max_depth 8 in three folds and 4 in two folds; max_features was sqrt and min_samples_leaf was 3 throughout. The final test model is separately tuned on the full training partition.

Raw and calibrated operating points

Model	Train OOF cutoff	Test FN	Test FP	Test cost / record	Approval rate
Raw Random Forest	0.22	8	77	0.585	35.5%
Platt-calibrated RF	0.17	8	75	0.575	36.5%

The calibrated model has its own cutoff, selected from cross-fitted calibrated training OOF probabilities. The raw cutoff must not be reused on calibrated scores. The small test-cost difference (0.010 per record) is not enough to establish a deployment winner.

Calibration diagnostics on the held-out test set

Metric	Raw RF	Platt calibrated RF	Preferred direction
ROC-AUC	0.809	0.809	Higher
Average precision	0.690	0.690	Higher
Brier score	0.161	0.153	Lower
Log loss	0.493	0.475	Lower
ECE (equal-width, 5 bins)	0.084	0.040	Lower
Calibration intercept	0.429	–0.082	Closer to 0
Calibration slope	1.724	1.037	Closer to 1

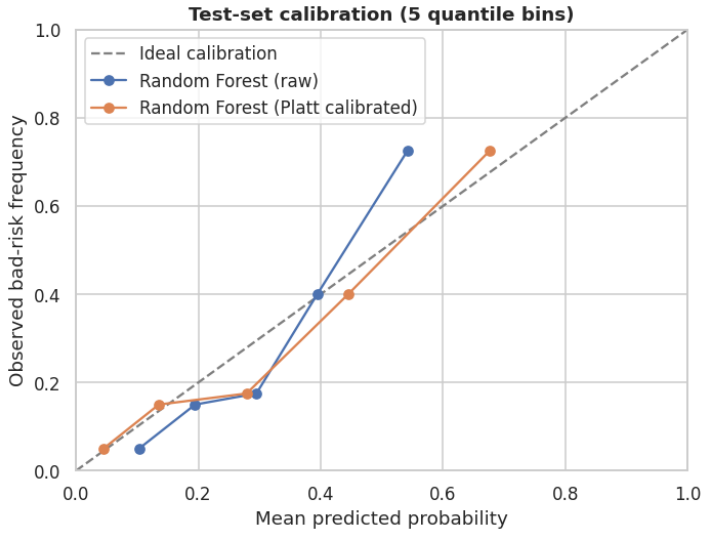


Figure 2. Test-set reliability curves using five quantile bins. Ranking metrics are unchanged by monotonic Platt scaling; probability error improves on this small test sample.

3. Model discrimination and explainability

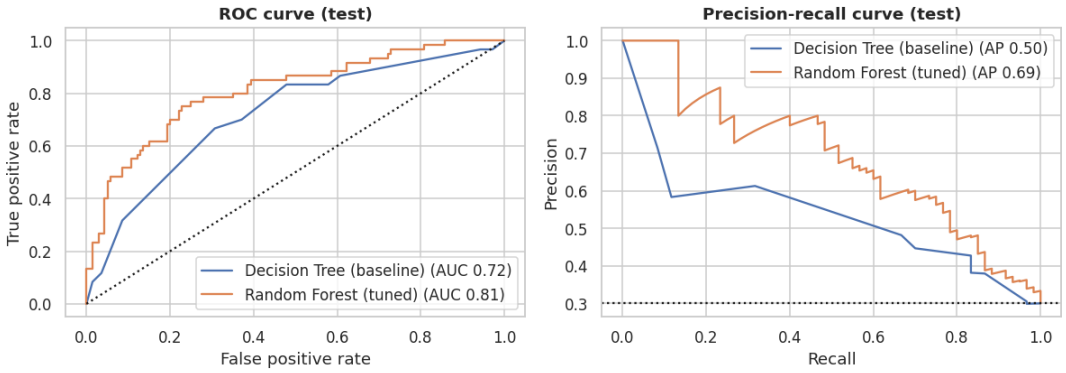


Figure 3. ROC and precision-recall curves on the full held-out test set.

SHAP global explanation

The notebook uses the maintained **shap** package directly: *shap.TreeExplainer* computes probability-space TreeSHAP values and *shap.summary_plot* generates both the mean-absolute-value bar plot and beeswarm. There is no custom SHAP substitute or package-unavailable fallback. Local additivity is checked against the fitted Random Forest probabilities. Global and local explanations use a fixed-seed stratified sample of 100 held-out test records (50 per class); headline test metrics use all 200 records.

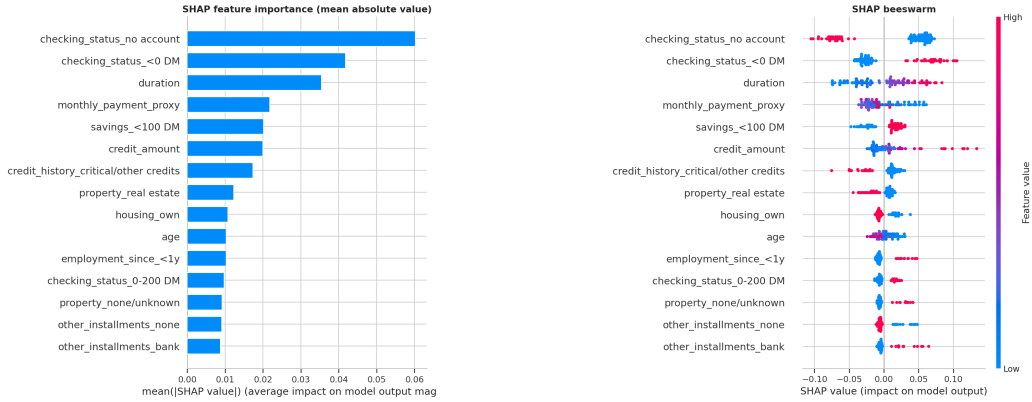


Figure 4. SHAP global bar and beeswarm plots on the stratified 100-record explanation sample. Checking-account status is the strongest original-feature driver, followed by duration and savings. SHAP impact size is not direction or causality.

4. Grouped importance, subgroup audit and limitations

Grouped original-feature importance

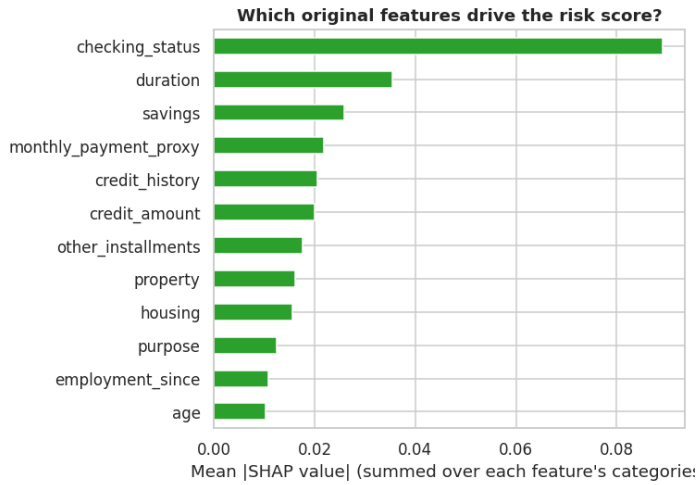


Figure 5. One-hot encoded SHAP contributions are summed back to original input fields before computing mean absolute importance.

Subgroup audit and responsible-use limits

Observed diagnostic	Interpretation / next step
Age 18–24: observed bad-risk label rate 48.7%; mean score 37.8%; approval 20.5%.	A 10.9 percentage-point calibration gap in this small subgroup warrants further review; it is not by itself a legal finding.
Age 35–49 approval rate: 49.2%.	Approval and error-rate differences require larger-sample, jurisdiction-specific fairness review.
personal_status_sex combines sex and marital status.	It cannot isolate sex effects. Removing it and foreign_worker from model inputs does not eliminate proxy risk.
1,000 historical records; test n=200; no timestamp or income/contractual instalment.	Obtain a larger recent portfolio, out-of-time data, affordability variables, and uncertainty intervals before operational use.

Conclusion. This is an educational benchmark, not a production underwriting model. The nested OOF and calibration changes reduce selection leakage in operating-point estimation, but small-sample uncertainty, selection bias, proxy effects and the absence of realised-default outcomes remain material limitations. SHAP and LIME are explanations of model behaviour, not causal evidence.

Sources: supplied Statlog German Credit archive and its german.doc documentation; UCI Machine Learning Repository, Statlog German Credit Data (ID 144).

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Thank you

With appreciation

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German Credit — Credit Risk Scoring Exercise